8. Monoalphabetic Substitution Cipher Using a Keyword

CODE:

```
plain = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
keyword = "CIPHER"

cipher = keyword

for ch in plain:
    if ch not in cipher:
        cipher += ch

print("Plain alphabet :", plain)
print("Cipher alphabet:", cipher)

message = input("Enter plaintext: ").upper()
encrypted = ""

for ch in message:
    if ch in plain:
        encrypted += cipher[plain.index(ch)]
    else:
        encrypted += ch

print("Encrypted text:", encrypted)

decrypted = ""

for ch in encrypted:
    if ch in cipher:
        decrypted += plain[cipher.index(ch)]
    else:
        decrypted += ch

print("Decrypted text:", decrypted)
```

OUTPUT:

